

SIMD-0532: Rotor Migration — Network Transition Procedure

Document Ref: SIMD-0532	Category: Standard Core	Repository Instance:
Title: Rotor Network Migration	Status: Final	proposals/0532-rotor-migration.md
Author: Richard Patterson	Date Created: June 25, 2026	

© 2026 Richard Patterson and RESEN. All Rights Reserved.

This document defines the cluster handshake protocols, dual-routing states, feature-gate dependencies, and circuit-breaker thresholds required to transition the live Solana Mainnet consensus cluster from legacy Turbine block replication to the new Rotor data dissemination protocol.

1. Executive Summary

SIMD-0532 provides the operational rollout framework and network orchestration blueprints for deploying **Rotor** across the decentralized validator ecosystem. Because Rotor alters live, high-throughput message routing structures at the consensus level, a sudden cutover presents catastrophic cluster liveness risks. This migration document defines a deterministic, phased rollout schedule leveraging gossip-advertised protocol capabilities, feature-gate dependencies, and dual-routing logic to isolate network degradation events and maintain stable transaction flow during software version upgrades.

2. Migration Timeline & Activation Phases

The cluster transition proceeds across four discrete milestones controlled via runtime feature-gate configurations. Advancing between milestones depends directly on total active stake-weighted validator participation profiles.

Phase	Phase Identifier	Stake Requirement	Subsystem Operational Behavior
Phase 0	`Inert Deployment`	0% (Feature Locked)	Upgraded nodes accept code paths but explicitly execute Turbine routing exclusively.
Phase 1	`Gossip Handshake`	> 33.3% Upgraded State	Nodes advertise capability bits via Gossip and map live baseline topology reliability matrices.
Phase 2	`Dual Coexistence`	> 66.6% Upgraded State	Leaders execute Rotor pathways for upgraded nodes; fallback loops handle legacy nodes.
Phase 3	`Hard Cutover`	> 95.0% Cluster Stability	Turbine code paths fully disabled. Rotor established as standard dissemination method.

3. Gossip Capability Handshake Logic

To dynamically evaluate cluster alignment without hardcoding specific version ranges, a new data element is added into the standardized validator contact info framework inside the gossip protocol layers. Upgraded clients populate bits within an atomic protocol metadata array bitfield:

```
// Protocol capability vector appended to contact structures
pub const GOSSIP_CAPABILITY_ROTOR_SUPPORTED: u32 = 1 << 0;
pub const GOSSIP_CAPABILITY_ROTOR_ACTIVE: u32    = 1 << 1;

pub struct ProtocolHandshakeExtension {
  pub extensions_bitfield: u32,
  pub maximum_negotiated_hop_limit: u8,
}
```

Validators process this payload continuously across gossip cycles. If a node discovers a down-level peer that does not populate `GOSSIP\_CAPABILITY\_ROTOR\_SUPPORTED`, it marks that peer's address within an isolated legacy transport index map inside the localized `Smart\_Sampler` engine.

4. Coexistence Network Topology (Dual Routing)

During the extended execution window of Phase 2, the cluster operates in a mixed transport state. Block leaders are required to construct data pipelines that support both old and new communication structures without doubling localized upstream bandwidth egress costs.

The leader divides its transmission sets explicitly: **Upgraded Peers** are addressed through a probabilistic Smart-Sampling distribution layer using QUIC streams mapped directly to Alpenglow slices. **Legacy Peers** are grouped into a temporary, synthetic standalone Turbine tree. Egress shreds into the legacy tree are throttled to ensure they match exactly the baseline requirements necessary for structural tree-hops, preventing starvation across legacy branches until they complete upgrade milestones.

5. Fallback and Emergency Deactivation Routines

Maintaining cluster safety under real-world stress requires deterministic automated circuit-breaker overrides. If a critical network error condition occurs during Phase 2 or Phase 3 execution windows, an atomic deactivation routine triggers immediately.

5.1 Operational Circuit-Breaker Thresholds

Trigger Parameter	Trigger Limit Value	Automated Mitigation Action Sequence
Slot Lag Multiplier	> 12 Slots over 3 epochs	Instantly down-grade local operations to Phase 1; freeze feature-gate updates.
Slice Completion Failure Rate	> 15% of active slots	Force full leader fallback loops to baseline Turbine transmission modes.
Gossip Delink Disconnects	> 40% node topology drops	Isolate affected nodes; restrict sample calculations to high-stake tiers.

In the event of an automated fallback trigger, any active state inside the `Smart\_Sampler` engine is preserved but bypasses active lookup execution paths. This ensures that historical metrics remain intact for diagnostics while the client safely relies on standard legacy deterministic tree generation models to protect system liveness.

6. Interconnected Reference Architecture Specs

SIMD Ref	Framework Title	Operational Domain Contract
SIMD-0530	Rotor	Core algorithmic principles, mathematical formulas, and baseline parameters.
SIMD-0531	Rotor-Anza	Low-level Rust engine design, memory models, and pipeline specifications.
SIMD-0532	Rotor Migration	Cluster consensus handshake sequences, feature-activation gates, and staging targets (This file).

7. Bibliography & Source Foundations

- Solana Improvement Document — SIMD-0530: Rotor Core Protocol Specification. Richard Patterson, 2026.
- Solana Improvement Document — SIMD-0531: Rotor-Anza Technical Implementation Specification. Richard Patterson, 2026.
- Solana Improvement Document — SIMD-0326: Alpenglow (Votor) Consensus Architecture Core Layouts.
- Solana Improvement Document — SIMD-0357: Alpenglow Validator Admission Ticket Specifications.
- Kniep, Sliwinski, Wattenhofer — *Solana Alpenglow Consensus: Increased Bandwidth, Reduced Latency v1.1*, 2025.

© 2026 Richard Patterson and RESEN. All Rights Reserved.

This technical framework document is authored by Richard Patterson and protected under standard copyright regimes. The SIMD-0532 specification text represents an integration and deployment roadmap submitted under the Solana SIMD development lifecycle. Any unapproved modifications, redistribution, or extraction of deployment logic without explicit alignment from the copyright owners is strictly prohibited.